Question 3

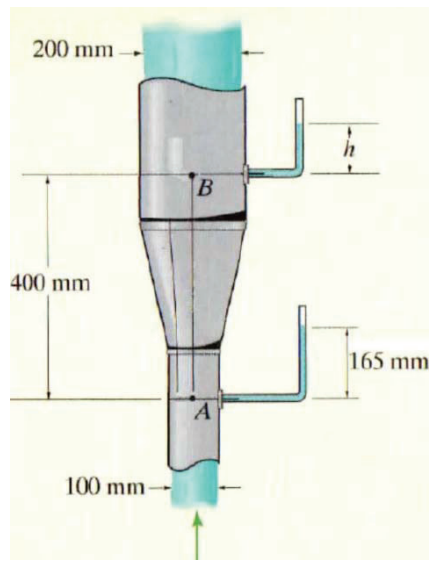
a.

Full form of the Bernoulli equation:

$$\frac{P_1}{\rho} + \alpha_1 \frac{v_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{v_2^2}{2} + gz_2$$

Where point 1 = point A and point 2 = point B.

Unit check: P/ρ [=] (N.m^3)/(m^2.kg) [=] (N.m)/kg × J/(N.m) [=] J/kg (energy per mass).



b.

First the pressure in point A should be calculated based on the information provided by the manometer, which is open to the atmosphere:

$$P_A = P_{atm} + \rho_{water} gh$$

$$P_A = 101325 + 1000 \times 9.81 \times 0.165 = 102\,944\ Pa\text{-a}$$

Now the other variables used in the Bernoulli equation should be calculated:

$$v_1 = \frac{Q}{A_1} = \frac{0.022\,\frac{\mathrm{m}^3}{\mathrm{s}}}{\frac{\pi\,(0.1\mathrm{m})^2}{4}} = 2.801\,\frac{m}{s}$$

$$v_2 = \frac{Q}{A_2} = \frac{0.022\,\frac{\mathrm{m}^3}{\mathrm{s}}}{\frac{\pi\,(0.2\mathrm{m})^2}{4}} = 0.700\ \frac{m}{s}$$

The Reynolds number should be calculated to identify if the flow is laminar or turbulent, so α can be found.

At the inlet:

$$Re = \frac{\rho\,\mathrm{V}_1\,\mathrm{D}_1}{\mu} = \frac{1000 \times 2.801 \times 0.1}{0.00151} = 185\,505$$

At the outlet:

$$Re = \frac{\rho V_2 D_2}{\mu} = \frac{1000 \times 0.700 \times 0.2}{0.00151} = 92\,750880$$

Therefore, the flow is turbulent in both segments of the pipe, so $\alpha_1 = \alpha_2 = 1$

Rearranging the Bernoulli equation and isolating for the pressure in point B:

$$P_2 = P_1 + \rho\left(\frac{v_1^2 - v_2^2}{2} + g(z_1 - z_2)\right)$$

$$P_2 = 102\,944 + 1000\left(\frac{2.801^2 - 0.700^2}{2} + 9.81 \times (0 - 0.4)\right) = 102\,697\ Pa\text{-a}$$

c.

Converting P_2 to height using the manometer equation:

$$P_2 = P_{atm} + \rho_{water} g h_2$$

$$h_2 = \frac{P_2 - P_{atm}}{\rho_{water} g} = \frac{102\,697 - 101\,325}{1000 \times 9.81} = 0.140\ m = \mathbf{140\ mm}$$